

Sarlink® TPV 3160

Teknor Apex Company - Thermoplastic Vulcanizate

Wednesday, April 10, 2024

General Information

Product Description

SARLINK® TPV 3100 series are engineered materials designed primarily for general purpose, automotive and industrial applications requiring a good balance of thermal, mechanical, and physical properties. SARLINK® 3160, available in NAT and BLK, is a medium hardness, low density, multi-purpose thermoplastic vulcanizate that can be processed by injection molding, blow molding or extrusion for applications such as grips, seals, gaskets, profiles, hose & tubes, bellows, and other articles.

General

Material Status	• Commercial: Active		
Availability	• Asia Pacific • Europe	• Latin America • North America	
Features	• Bondability • Chemical Resistant • General Purpose • Good Adhesion • Good Flexibility	• Good Moldability • Good Processability • Good Surface Finish • Good Weather Resistance • High Elasticity	• Low Density • Medium Hardness • Medium Heat Resistance • Resilient
Uses	• Automotive Applications • Automotive Exterior Parts • Automotive Interior Parts • Automotive Under the Hood • Diaphragms	• Gaskets • General Purpose • Industrial Applications • O-rings • Pipe Seals	• Plugs • Profiles • Rubber Replacement • Seals • Weatherstripping
Agency Ratings	• UL 94	• UL QMFZ2	• UL QMFZ8
RoHS Compliance	• RoHS Compliant		
Automotive Specifications	• BMW Unspecified Color: Black • CHRYSLER MS-AR-80 Type B Color: Black • CHRYSLER MS-AR-80 Type B Color: Natural • DAIMLER DBL 5562.30 Color: Black • FORD WSD-M2D379-A1 Color: Black • GM QK 003521 Color: Black • GM QK 003521 Color: Natural • HONDA Unspecified Color: Black • PSA Peugeot-Citroën B62 0300 version G Color: Black • SAE J3000 Color: Black • SAE J3000 Color: Natural		
UL File Number	• QMFZ2.E54709		
Appearance	• Black	• Natural Color	• Opaque
Forms	• Pellets		
Processing Method	• Blow Molding	• Extrusion	• Injection Molding

ASTM & ISO Properties ¹

Physical	Nominal Value	Unit	Test Method
Density / Specific Gravity	0.950		ASTM D792
Density	0.950	g/cm ³	ISO 1183
Elastomers	Nominal Value	Unit	Test Method
Tensile Stress			ASTM D412
Across Flow : 100% Strain	363	psi	
Flow : 100% Strain	551	psi	
Tensile Stress			ISO 37
Across Flow : 100% Strain	363	psi	
Flow : 100% Strain	551	psi	

Revision Date: 4/9/2018

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Elastomers	Nominal Value	Unit	Test Method
Tensile Strength			ASTM D412
Across Flow : Break	914	psi	
Flow : Break	783	psi	
Tensile Stress			ISO 37
Across Flow : Break	914	psi	
Flow : Break	783	psi	
Tensile Elongation			ASTM D412
Across Flow : Break	640	%	
Flow : Break	270	%	
Tensile Elongation			ISO 37
Across Flow : Break	640	%	
Flow : Break	270	%	
Tear Strength - Across Flow	180	lbf/in	ASTM D624
Tear Strength - Across Flow ²	183	lbf/in	ISO 34-1
Compression Set			ASTM D395
73°F, 22 hr	23	%	
158°F, 22 hr	34	%	
257°F, 70 hr	55	%	
Compression Set			ISO 815
73°F, 22 hr	23	%	
158°F, 22 hr	34	%	
257°F, 70 hr	55	%	
Hardness	Nominal Value	Unit	Test Method
Durometer Hardness			ASTM D2240
Shore A, 5 sec, Extruded	62		
Shore A, 5 sec, Injection Molded	65		
Shore Hardness			ISO 868
Shore A, 5 sec, Extruded	62		
Shore A, 5 sec, Injection Molded	65		
Thermal	Nominal Value	Unit	Test Method
RTI Elec	122	°F	UL 746B
RTI Imp	122	°F	UL 746B
RTI Str	122	°F	UL 746B
Aging	Nominal Value	Unit	Test Method
Change in Tensile Strength in Air - Across Flow			ASTM D573
275°F, 1000 hr	-4.0	%	
100% Strain, 275°F, 1000 hr	3.0	%	
302°F, 168 hr	-1.0	%	
100% Strain, 302°F, 168 hr	7.0	%	
Change in Tensile Strength in Air - Across Flow			ISO 188
275°F, 1000 hr	-4.0	%	
100% Strain 275°F, 1000 hr	3.0	%	
302°F, 168 hr	-1.0	%	
100% Strain 302°F, 168 hr	7.0	%	
Change in Ultimate Elongation in Air - Across Flow			ASTM D573
275°F, 1000 hr	-5.0	%	
302°F, 168 hr	-11	%	

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Aging	Nominal Value	Unit	Test Method
Change in Tensile Strain at Break in Air - Across Flow			ISO 188
275°F, 1000 hr	-5.0	%	
302°F, 168 hr	-11	%	
Change in Durometer Hardness in Air			ASTM D573
Shore A, 275°F, 1000 hr	2.0		
Shore A, 302°F, 168 hr	3.0		
Change in Shore Hardness in Air			ISO 188
Shore A, 275°F, 1000 hr	2.0		
Shore A, 302°F, 168 hr	3.0		
Change in Volume (257°F, 70 hr, in IRM 903 Oil)	120	%	ASTM D471
Change in Volume (257°F, 70 hr, in IRM 903 Oil)	120	%	ISO 1817
Electrical	Nominal Value	Unit	Test Method
Volume Resistivity	1.0E+16	ohms·cm	ASTM D257
Flammability	Nominal Value	Unit	Test Method
Flame Rating (0.06 in, Natural and Black Colors)	HB		UL 94
Additional Information	Nominal Value	Unit	Test Method
Apparent Shear Viscosity - Capillary, @ 206/s			
392°F	310	Pa·s	ASTM D3835
392°F	310		ISO 11443

Legal Statement

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Processing Information		
Injection	Nominal Value	Unit
Drying Temperature	180	°F
Drying Time	3.0	hr
Rear Temperature	356 to 419	°F
Middle Temperature	356 to 419	°F
Front Temperature	356 to 419	°F
Nozzle Temperature	369 to 428	°F
Processing (Melt) Temp	365 to 428	°F
Mold Temperature	50 to 131	°F
Back Pressure	14.5 to 145	psi
Screw Speed	100 to 200	rpm
Extrusion	Nominal Value	Unit
Drying Temperature	180	°F
Drying Time	3.0	hr
Cylinder Zone 1 Temp.	356 to 392	°F
Cylinder Zone 2 Temp.	356 to 401	°F
Cylinder Zone 3 Temp.	369 to 410	°F
Cylinder Zone 4 Temp.	369 to 410	°F
Melt Temperature	383 to 419	°F
Die Temperature	383 to 419	°F

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Extrusion	Nominal Value	Unit
Take-Off Roll	68 to 122	°F

Extrusion Notes
Screen Pack: 20 to 60 mesh
Screw: general purpose
Compression Ratio: 3:1

Notes

- ¹ Typical properties: these are not to be construed as specifications.
- ² Method Ba, Angle (Unnicked)

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